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MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION APRIL 2026

SECOND SEMESTER

Discipline Specific Core Course (DSC) - MG2DSCZIM101 - PUBLIC HEALTH MICROBIOLOGY

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(1) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A
Fill in the blanks
Answer all questions
Each question carries 1 marks

1. ETO stands for..... **[K]/[CO4]**

Answer: Ethylene Oxide

2. CFU Stands for …… **[K]/[CO2]**

Answer: Colony Forming Unit

3…… is called as public health. **[U]/[CO1]**

Answer: Sanitation

4…….. is the gram-negative bacteria in waste water. **[U]/[CO4]**

Answer: Escherichia coli

5. Water can act as a…… of transmission for various infections **[K]/[CO4]**

Answer: Medium

6. DTP stands for …… **[U]/[CO3]**

Answer: Dipthria, tetanus and pertussis

7. HEPA Filter remove particles as small as ……. diameter **[K]/[CO2]**

Answer: 0.3m

8. Molds and yeasts are eliminated during..... **[K]/[CO5]**

Answer: Sterilization

9. In between 1817-1923 there were six…….pandemics recorded **[U]/[CO4]**

Answer: cholera

10. ……are a significant mode of transmission for respiratory infections **[U]/[CO4]**

Answer: Droplets

[1x10 = 10]

Part B
Short Answer Questions
Answer any 10 questions
Each question carries 2 marks

11.Outline Air Pollutants **[K]/[CO1]**

Major air pollutants consist of Particulate Matter (PM2.5/PM10), Sulfur Dioxide (SO2), and Nitrogen Oxides (NOx), which are primarily released from vehicle exhausts and industrial factories. Carbon Monoxide (CO) and Ground-level Ozone (O3) are also critical pollutants that trigger respiratory issues and contribute to environmental haze.

12. Classify bacterial waterborne diseases. **[U]/[CO4]**

Bacterial waterborne diseases are typically classified by the type of pathogen and its clinical manifestation:

Enteric Infections: Primarily affecting the digestive tract, such as Cholera (Vibrio cholerae) and Typhoid Fever (Salmonella Typhi).

Bacillary Infections: Causes acute inflammation of the colon, most notably Bacillary Dysentery (Shigella species)

13. Classify the immunization of food borne infections. **[U]/[CO5]**

Immunization against foodborne pathogens involves both active and passive strategies to prevent outbreaks:

Vaccination: Targeted shots for specific high-risk pathogens like Hepatitis A and Typhoid Fever to build long-term immunity.

Passive Immunity: Use of pre-formed antibodies (immunoglobulins) in post-exposure scenarios to provide immediate, short-term protection.

14. What is acid rain? **[K]/[CO1]**

Acid rain refers to any form of precipitation with acidic components, typically resulting from the reaction of Sulfur Dioxide (SO2) and Nitrogen Oxides (NOx) with water, oxygen, and other chemicals in the atmosphere. It results in a pH level lower than the natural average of 5.6, causing damage to ecosystems and infrastructure.

15. Explain the types of food hygiene. **[U]/[C05]**

Food hygiene is categorized into four main pillars to ensure safety from farm to table:Personal Hygiene: Proper handwashing and health standards for food handlers.

Environmental Hygiene: Maintaining clean facilities, equipment, and effective pest control.

Food Handling Hygiene: Safe storage (temperature control), preparation (preventing cross-contamination), and thorough cooking.

16. Discuss high temp water treatment. **[U]/[CO4]**

This process uses heat to eliminate pathogenic microorganisms, typically through two methods:

Boiling: Heating water to 100°C for at least one minute to kill most bacteria, viruses, and protozoa.

Pasteurization: Heating water to a specific temperature (e.g., 72°C) for a set duration to reduce the microbial load without boiling, often used in large-scale solar water disinfection.

17. Explain enzymatic spoilage. [K]/[CO5]

Enzymatic spoilage occurs when natural enzymes within food remain active after harvest or slaughter, breaking down cellular structures. This leads to browning (like in sliced apples), softening of tissues, and the development of off-flavors, even in the absence of microbial growth.

18. Write about solid waste management. [K]/[CO5]

This is the systematic control of the generation, storage, collection, transport, processing, and disposal of solid waste. It prioritizes the "3 Rs" (Reduce, Reuse, Recycle) and utilizes methods like composting, incineration, and sanitary landfills to minimize environmental and health impacts.

19. Describe staphylococcal food poisioning. [K]/[CO5]

Caused by ingesting toxins produced by Staphylococcus aureus, this illness is characterized by a very rapid onset (usually 30 minutes to 8 hours). It typically stems from improper handling of "ready-to-eat" foods, as the bacteria can multiply and produce heat-stable toxins if food is left at room temperature.

20. Discuss the symptoms of bacillary dystentry. [U]/[CO5]

Caused by Shigella bacteria, the primary symptoms include:

Severe Mucoid Diarrhea: Frequent stools often containing blood and mucus.

Abdominal Cramping: Intense pain and "tenesmus" (a constant feeling of needing to pass stool).

Systemic Signs: High fever, nausea, and dehydration.

21. Explain morphology of Hepatitis A Virus. [U]/[CO4]

The Hepatitis A virus is a small, non-enveloped virus belonging to the Picornaviridae family. It features an icosahedral capsid (roughly spherical) and contains a single-stranded RNA genome, making it highly resistant to environmental stressors like heat and acid.

22. Examine the health risk of air pollution [U]/[CO2]

Air pollution impacts multiple body systems, leading to both acute and chronic conditions:

Respiratory Issues: Aggravation of asthma, chronic bronchitis, and increased risk of lung cancer.

Cardiovascular Strain: Higher incidence of heart attacks and strokes due to systemic inflammation caused by fine particulate matter entering the bloodstream.

[2x10 = 20]

Part C

Short Essay Questions

Answer any **5** questions

Each question carries **4** marks

23. Discuss industrial waste disposal lead to water contamination. [U]/[CO4] (Mechanism of contamination – 2 marks, Industrial waste concept – 2 marks).

Industries often discharge untreated effluents containing heavy metals, toxic chemicals, and organic solvents directly into water bodies. These pollutants alter the chemical composition of the water, making it toxic for aquatic

life and unsafe for human consumption. Seepage from industrial landfills can also leak into groundwater, leading to long-term contamination of drinking water sources.

24. Describe methods to detect egg spoilage. [U]/[CO5] (Any four methods 1 mark each)

The Float Test is a simple method where fresh eggs sink in water while spoiled eggs float due to increased air cells. Candling involves holding the egg against a bright light to check for internal spots, blood rings, or a displaced yolk. Sensory evaluation can detect spoilage through the presence of a "rotten egg" sulfur smell or a thin, watery consistency of the whites.

25. Investigate the causes of Waste Pollution. [U]/[CO1] (Any four causes of waste pollution 1 mark each)

Waste pollution is primarily caused by rapid urbanization and the over-consumption of non-biodegradable materials like plastics and electronics. Inadequate waste collection systems and the lack of public awareness regarding segregation lead to littering and illegal dumping. Industrial expansion and poor management of hazardous materials further contribute to the accumulation of toxic waste in the environment

26. Classify hospital waste. [K]/[CO5] (Any 4 categories with examples – 4 marks)

Hospital waste is classified into Infectious Waste (swabs, bandages, cultures) and Pathological Waste (body parts, blood, fluids). It also includes Sharps (needles, scalpels), Pharmaceutical Waste (expired medicines), and General Waste (paper, food packaging). Radioactive waste and chemical waste from labs are also categorized separately due to their specific disposal requirements.

27. State UV light. [K]/[CO4] (Definition – 1 mark, Properties – 2 marks, Uses – 1 mark)

Ultraviolet (UV) light is a form of electromagnetic radiation that acts as a powerful germicidal agent by damaging the DNA of microorganisms. When pathogens are exposed to UV-C rays, they lose their ability to reproduce and cause infections. It is widely used for sterilizing laboratory surfaces, treating drinking water, and purifying air in medical facilities.

28. Distinguish between influenza and tuberculosis. [K]/[CO3] (Any 2 points on Influenza – 2 marks, Any 2 points on Tuberculosis – 2 marks)

Influenza is a viral respiratory infection characterized by sudden onset, high fever, and body aches, usually lasting for a week. Tuberculosis is a chronic bacterial infection caused by Mycobacterium tuberculosis that primarily affects the lungs over a long period. While the flu is prevented by seasonal vaccines, TB requires a long-term course of specialized antibiotics and has a much slower progression**.**

29. Illustrate factors affecting microbial growth. [U]/[CO5] (Any four factors with brief explanation example: Temperature, pH, Oxygen, Nutrient availability, Moisture – 4 marks)

Microbial growth is heavily influenced by Temperature, where most pathogens thrive between 5°Cand 60°C (the danger zone). Moisture levels, or water activity, are essential for metabolic processes and cellular reproduction in bacteria and fungi. The pH Level of the environment dictates growth, as most bacteria prefer neutral conditions while molds can tolerate acidity. Lastly, the availability of Nutrients like carbon and nitrogen provides the energy necessary for microbes to multiply

[4x5 = 20]